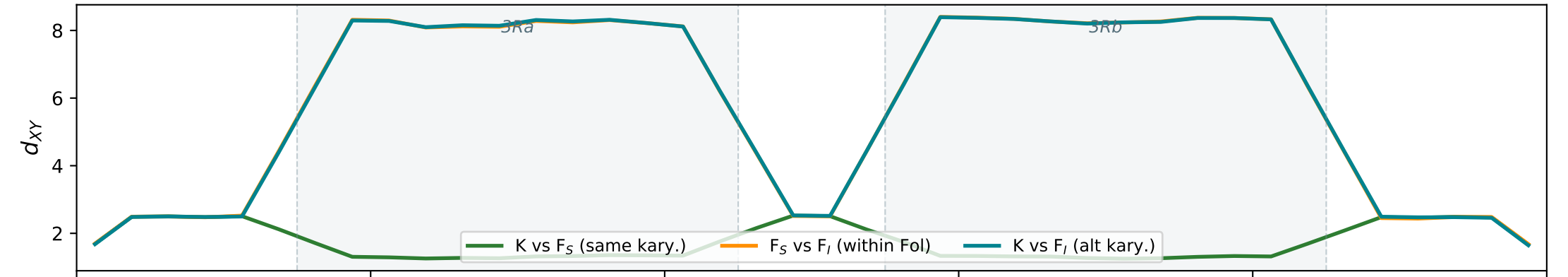
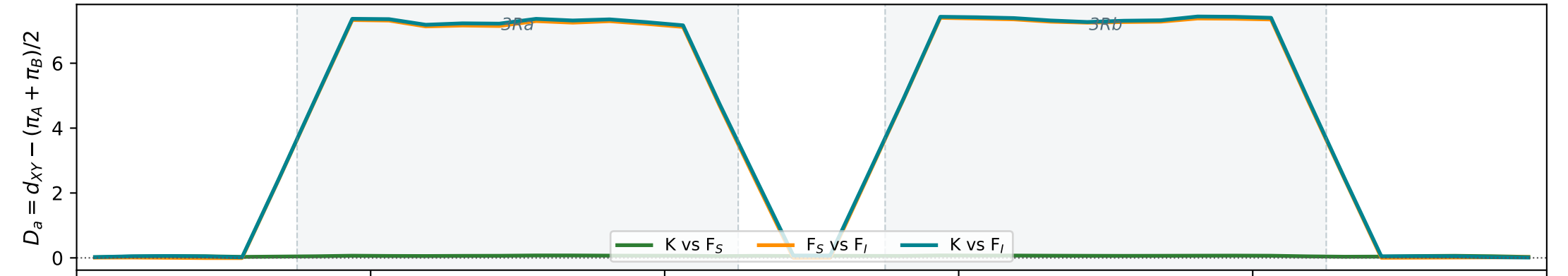


An. funestus: Kiribina vs Folonzo divergence at 3Ra + 3Rb inversions
msinv coalescent simulation with real demographic parameters

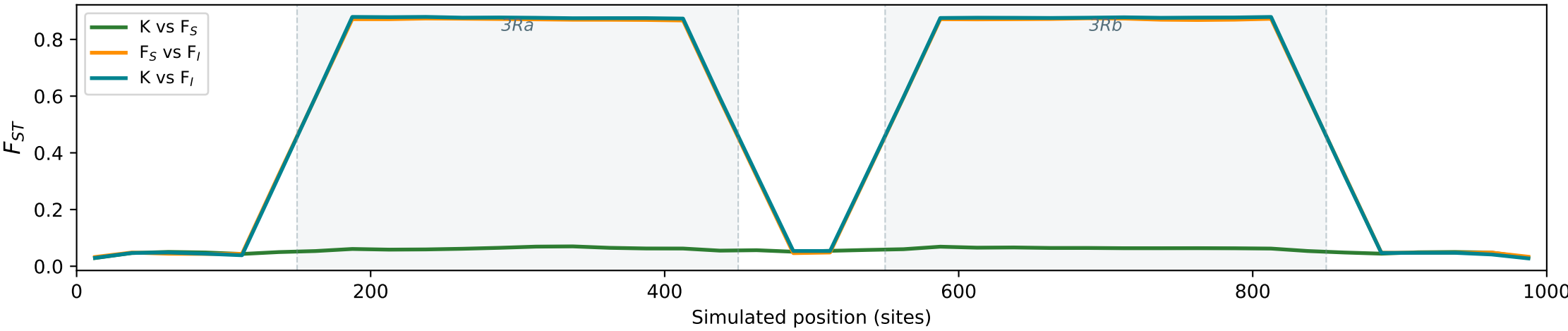
A. Raw absolute divergence d_{XY} — swamped by π_{Fol} with $Ne_F=3M$ (shared π dominates)



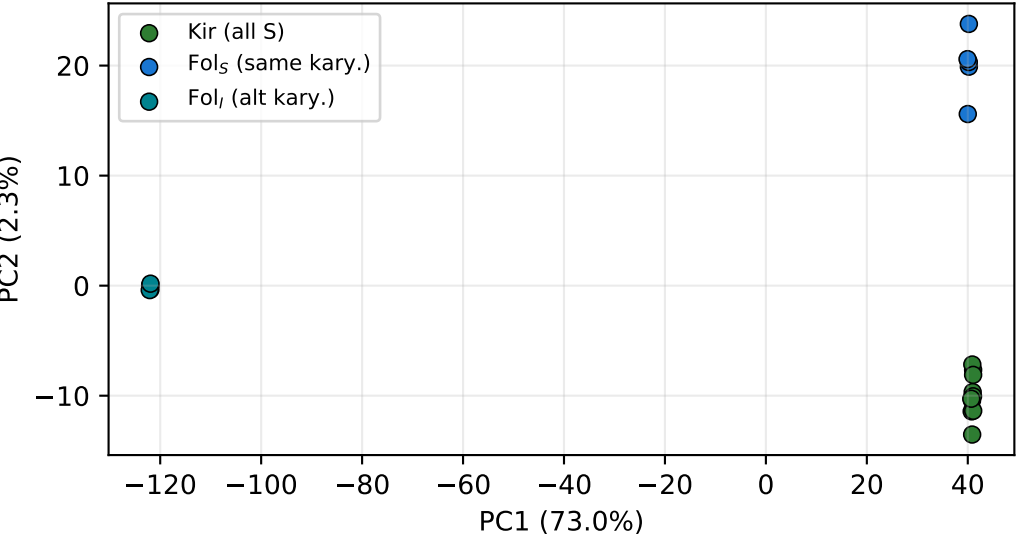
B. Net divergence D_a (Nei 1987; subtracts shared π) — isolates the inversion signal



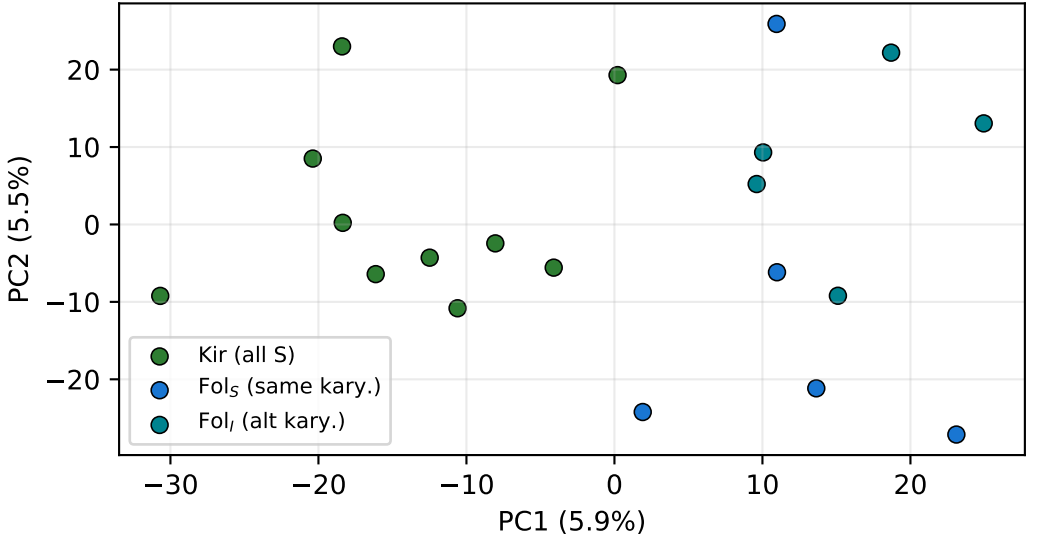
C. Relative divergence F_{ST}



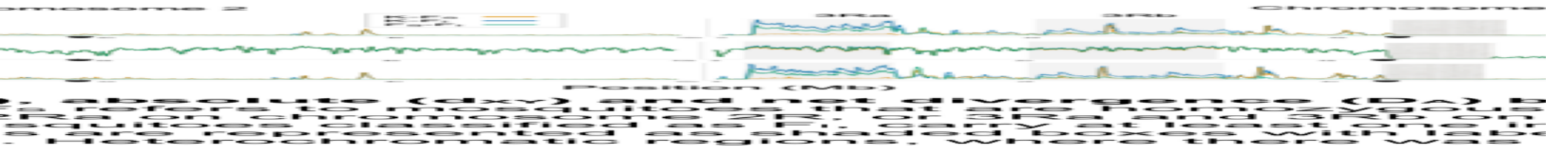
D1. PCA INSIDE 3Ra inversion — karyotype clustering



D2. PCA OUTSIDE inversions — no karyotype structure



E. Empirical divergence, chromosome 3 (Small et al. 2023, Fig. S13)



Parameters (Table S8): $Ne_K=70,000$, $Ne_F=3,000,000$, $Ne_{Anc}=44,000$ — EXTREME Ne asymmetry (ratio $\sim 43\times$)
 $T_{split}=14,000$ gen ($\sim 1,300$ yr), $T_{inv}\sim 385,000$ gen (~ 35 kyr). $\theta=62.5$, $\rho=704$ (100 kb). Gene flux $\gamma=0$ (recombination modifier only). 300 replicates.
KEY INSIGHT: d_{XY} is swamped by high π_{Fol} . D_a (net divergence) recovers the inversion signal that matches empirical F_{ST} and PCA patterns.